Learning and Efficiency in the Market for Physician Referrals

Ian McCarthy
Emory University & NBER

Seth Richards-Shubik
Johns Hopkins University & NBER

SEA Annual Meeting, November 2026

Experts who guide buyers must learn about quality themselves

- Buyers of complex services often rely on an expert intermediary
  - Financial advisors, staffing agencies, real estate agents
- The intermediary is imperfectly informed too, and learns about quality from the outcomes it observes
- PCP referrals are a leading case
  - A PCP repeatedly sends patients to surgeons whose quality is hard to observe
  - Quality is revealed only gradually, through the outcomes of the PCP's own patients

How well do PCPs learn about specialist quality, and how much does it matter for where patients go and how they fare?

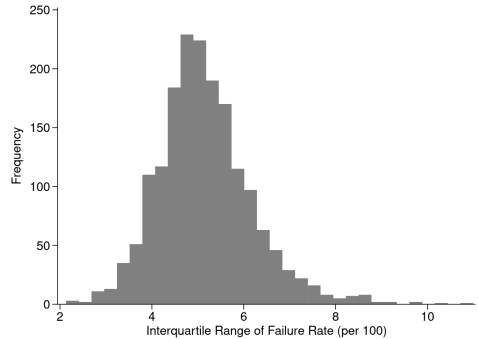
Setting and Data

Setting and data: joint replacement in Medicare

- Planned, elective major joint replacements, a predictable pathway where patients rely on a PCP referral; about 1.5 million a year in the US, 10% with a complication or readmission
- All Medicare fee-for-service claims, 2008–2018: 4.5 million surgeries
  - Referring PCP is the one with the most office visits in the year before surgery (2+ visits); identifies a PCP for about 80% of procedures
  - PCPs with 20+ referrals and surgeons with 20+ surgeries a year: just over 2 million referrals, 51,067 PCPs, 9,254 surgeons
- Quality is a failure within 90 days: death, readmission, or complication, following CMS's Comprehensive Care for Joint Replacement model
  - 9 failures per 100 surgeries on average
- Traditional Medicare has no networks, so any surgeon in the market is available

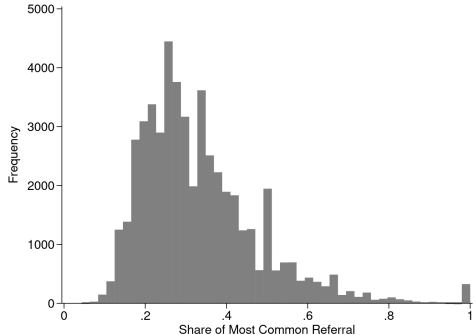
Surgeon quality varies within markets, and PCPs use only a few surgeons

- The 25th and 75th percentile surgeons in a market differ by 3.97 to 6.43 failures per 100, against a mean of 9



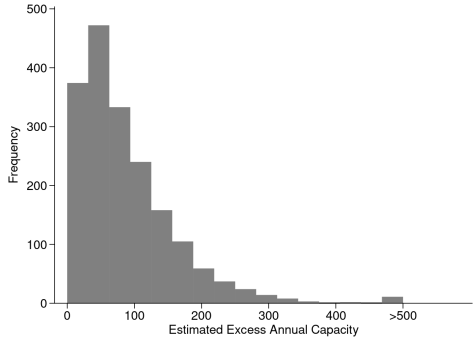
Surgeon quality varies within markets, and PCPs use only a few surgeons

- The 25th and 75th percentile surgeons in a market differ by 3.97 to 6.43 failures per 100, against a mean of 9
- A PCP uses about 3.11 of the roughly 69 surgeons in the market each year, and the most-used surgeon gets a third of a PCP's referrals



Surgeon quality varies within markets, and PCPs use only a few surgeons

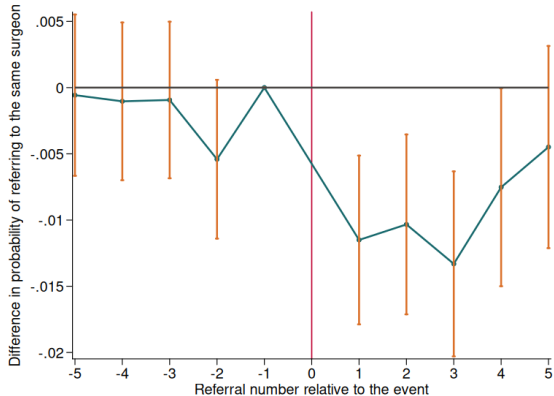
- The 25th and 75th percentile surgeons in a market differ by 3.97 to 6.43 failures per 100, against a mean of 9
- A PCP uses about 3.11 of the roughly 69 surgeons in the market each year, and the most-used surgeon gets a third of a PCP's referrals
- In 39% of market-years, top-quartile surgeons appear to have room for the bottom quartile's patients



Do PCPs Respond to Bad Outcomes?

PCPs refer less to a surgeon after their own patient has a bad outcome

- Every referral is an event; the outcome is whether the PCP's *next* referral goes to the same surgeon
- Failures against successes, with surgeon-by-year fixed effects and the pair's prior history
- A failure lowers the chance of returning by -6.8% (-0.0110 on a base of 0.160), unchanged with PCP fixed effects
- Largest where the PCP knows the surgeon least: -21.2% at a first referral, -3.3% after five or more



Event study in referral order, failure minus success windows

A Learning Model of Referrals

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

belief about quality, with weight α

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

belief about quality, with weight α

distance to the surgeon

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

- $\alpha\, m_{ijt}$: belief about quality, with weight α
- $\pi\, x_{ijt}$: distance to the surgeon
- $f(e_{ijt})$: familiarity: past referrals to j

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

- $\alpha\, m_{ijt}$: belief about quality, with weight α
- $\pi\, x_{ijt}$: distance to the surgeon
- $f(e_{ijt})$: familiarity: past referrals to j
- $\gamma\, n_{jt}$: congestion: the surgeon's other patients

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

- $\alpha\, m_{ijt}$: belief about quality, with weight α
- $\pi\, x_{ijt}$: distance to the surgeon
- $f(e_{ijt})$: familiarity: past referrals to j
- $\gamma\, n_{jt}$: congestion: the surgeon's other patients
- ξ_j: surgeon fixed effect

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

- $\alpha\, m_{ijt}$: belief about quality, with weight α
- $\pi\, x_{ijt}$: distance to the surgeon
- $f(e_{ijt})$: familiarity: past referrals to j
- $\gamma\, n_{jt}$: congestion: the surgeon's other patients
- ξ_j: surgeon fixed effect

Learning

Beliefs start from a common beta prior and update with each patient's outcome

$$m_{ijt} = \frac{\rho\eta + \text{past successes}}{\eta + \text{past referrals}}$$

ρ = prior mean, η = prior strength

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \alpha\, m_{ijt} + \pi\, x_{ijt} + f(e_{ijt}) + \gamma\, n_{jt} + \xi_j + \epsilon_{ijt}$$

- $\alpha\, m_{ijt}$: belief about quality, with weight α
- $\pi\, x_{ijt}$: distance to the surgeon
- $f(e_{ijt})$: familiarity: past referrals to j
- $\gamma\, n_{jt}$: congestion: the surgeon's other patients
- ξ_j: surgeon fixed effect

Learning

Beliefs start from a common beta prior and update with each patient's outcome

$$m_{ijt} = \frac{\rho\eta + \text{past successes}}{\eta + \text{past referrals}}$$

ρ = prior mean, η = prior strength

Behavior

Myopic PCPs refer on current beliefs

Forward-looking PCPs also value what a referral teaches them, through a Gittins index

PCPs learn about surgeon quality from their own patients' outcomes

$$U_{ijt} = \underbrace{\alpha\, m_{ijt}}_{\text{belief about quality, with weight } \alpha} + \underbrace{\pi\, x_{ijt}}_{\text{distance to the surgeon}} + \underbrace{f(e_{ijt})}_{\text{familiarity: past referrals to } j} + \underbrace{\gamma\, n_{jt}}_{\text{congestion: the surgeon's other patients}} + \underbrace{\xi_j}_{\text{surgeon fixed effect}} + \epsilon_{ijt}$$

Learning

Beliefs start from a common beta prior and update with each patient's outcome

$$m_{ijt} = \frac{\rho\eta + \text{past successes}}{\eta + \text{past referrals}}$$

ρ = prior mean, η = prior strength

Behavior

Myopic PCPs refer on current beliefs

Forward-looking PCPs also value what a referral teaches them, through a Gittins index

Estimation

Logit in each of 306 HRRs

Congestion by 2SLS, distance-based instrument

ρ = market success rate, η = 1 or 5

Identification in one slide

The belief term splits into the prior and the past success rate, $\bar{y}$:

$$\alpha\, m_{ijt} = \alpha\, \frac{\rho\eta}{\eta + e_{ijt}} + \alpha\, \frac{e_{ijt}}{\eta + e_{ijt}}\, \bar{y}_{ijt}$$

- α: a PCP with many past patients at two surgeons but different success rates; as $e \to \infty$ the belief term is $\alpha\bar{y}$
- η: how fast the effect of the success rate grows with the number of past patients; weakly identified, so fixed at 1 or 5
- Familiarity $f(e)$: the effect of past referrals that remains once the belief term is accounted for
- Congestion γ: surgeon-by-period fixed effects, then 2SLS with volume predicted from *other* patients' distances

Estimates

		Percentile across HRRs				
Myopic model	Mean	10th	25th	50th	75th	90th
α ($\eta = 1$)	0.202	0.000	0.000	0.117	0.283	0.537
α ($\eta = 5$)	0.397	0.000	0.000	0.212	0.561	1.089
π (distance, per mile)	-0.069	-0.099	-0.082	-0.068	-0.053	-0.038

- α/π: a success is worth about 2.91 miles of travel, in line with small quality elasticities elsewhere
- Familiarity is large and rising: 1.1 after one past patient, 2.4 after eight to ten
- Congestion: 100 more patients over three years lowers the referral probability by 7–9%
- Forward-looking model: α somewhat larger (0.26 at $\eta = 1$), everything else the same

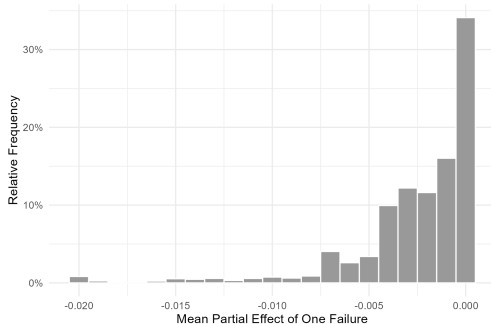
Counterfactuals

What we find

PCPs weigh quality lightly relative to other factors

0 to 2 pp

drop in referral probability after one more failure, across markets. Where PCPs respond, the average is 0.32 pp.



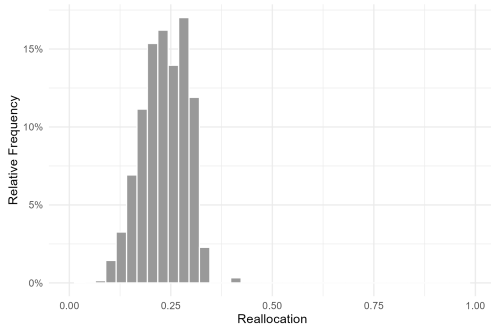
What we find

0 to 2 pp PCPs weigh quality lightly relative to other factors

Full information

23.25%

of patients go to a different surgeon when PCPs know true quality and keep their existing relationships. Failures fall by 0.07 per 100, with gains in most markets



What we find

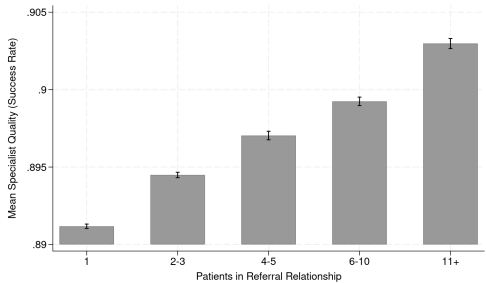
0 to 2 pp PCPs weigh quality lightly relative to other factors
23% reallocated under full information, with existing relationships

Existing relationships are with better surgeons

31.75%

of patients move under full information when the relationships are reset, with no health gain

Success rates rise with relationship length, so familiarity has already sorted PCPs toward better surgeons



What we find

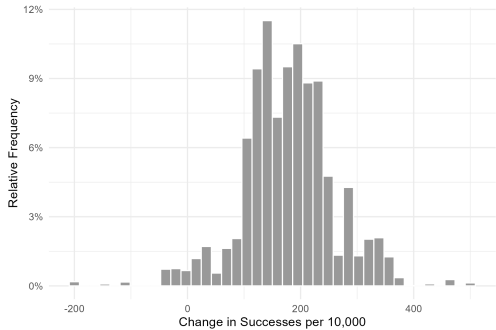
0 to 2 pp PCPs weigh quality lightly relative to other factors
23% reallocated under full information, with existing relationships
32% reallocated when the relationships are reset, with no health gain

Quality weighted like distance

20%

of failures avoided when one SD of quality counts as much as one SD of distance

150 to 200 more successes per 10,000, still subject to distance and surgeon capacity



What we find

0 to 2 pp	PCPs weigh quality lightly relative to other factors
23%	reallocated under full information, with existing relationships
32%	reallocated when the relationships are reset, with no health gain
20%	of failures avoided when quality counts like distance

Takeaway

- The gains from quality information alone are bounded by the small weight PCPs place on quality, and they depend on the referral relationships PCPs already have
- Tools that raise that weight could deliver substantially larger health improvements

Summary

- Surgeon quality varies widely within markets, and PCPs respond to their own patients' outcomes, most when they know the surgeon least
- A structural learning model of referrals with habit persistence and capacity constraints puts a small weight on quality: a success is worth about 2.91 miles
- Full information reallocates about a quarter of referrals with small but meaningful health gains, and the gains depend on the referral relationships PCPs already have
- Quality information works best alongside existing referral relationships rather than in place of them

ian.mccarthy@emory.edu

Backup

Re-referral design

- For the referral of patient t from PCP i to surgeon j, with outcome Y_{ijt} and prior history e_{ijt}, the outcome is $R_{ijt'} = 1$ if the PCP's next patient t' goes to j:

$$R_{ijt'} = \gamma_{j,y(t)} + \beta(1 - Y_{ijt}) + \sum_{h=1}^{5} \kappa_h \mathbf{1}\left(\min(e_{ijt}, 5) = h\right) + \varepsilon_{ijt}$$

- 1,190,521 referrals, 112,871 failures; $\hat{\beta} = -0.0110$ (s.e. 0.0011); standard errors clustered by surgeon and by PCP
- Event study in referral order: a window is the PCP's five referrals before and after a given referral; failure windows have their only failure at position 0, success windows none

$$R_{ijt} = \mu_{ij} + \lambda_{jy,t} + \phi_{i,t} + \sum_{k \neq -1} \delta_k \mathbf{1}(t = k)(1 - Y_{ij0}) + \varepsilon_{ijt}$$

- First referral after a failure: -0.0115 (s.e. 0.0033) on a base of 0.153; the gap lasts through the third referral and fades by the fifth

Setting the priors

- Prior mean $\rho = a_0/(a_0 + b_0)$ equals the market's average success rate (about 0.90), as in Coscelli and Shum (2004), Crawford and Shum (2005), and Dickstein (2018)
- Prior strength $\eta = a_0 + b_0$ acts like a number of patients seen before the first real one
  - The likelihood is flat in η in simulations, so it is fixed at 1 or 5; the cross-partial that identifies it moves only from 0.033 to 0.056 as η goes from 1 to 20
  - PCPs refer about 4.40 patients a year, so these values bracket a sensible range
  - Other estimates and the counterfactuals are similar at both values; α scales with η

Congestion stands in for surgeon capacity

- $\gamma\, n_{jt}$: a surgeon treating more patients is less attractive for the next one (for example, wait times)
- Two steps (Richards-Shubik, Roberts and Donohue 2022, following Bayer and Timmins 2007)
  - Estimate the logit with a surgeon-by-period fixed effect $\delta_{j\tau} = \gamma n_{j\tau} + \xi_j$
  - Regress $\hat{\delta}_{j\tau}$ on $n_{j\tau}$ by 2SLS
- Instrument: the surgeon's patient count predicted from distances alone; other patients' distances shift the surgeon's volume but not this patient's referral (first-stage F above 1,000)
- In the counterfactuals, volumes and referrals are re-solved together as a fixed point

Forward-looking PCPs also value experimentation

- Myopic: refer to the surgeon with the highest expected payoff for this patient
- Forward-looking: also value what the referral teaches about the surgeon, for future patients
  - Gittins index $g(m, v)$, rising in the variance of beliefs, with the closed-form approximation of Brezzi and Lai (2002)
  - No parameters beyond the myopic model
- The forward-looking α is somewhat larger (0.26 vs. 0.20 at $\eta = 1$); counterfactual results are similar in the two versions

Risk adjustment changes little

	Mean $\hat{\alpha}$
• Baseline	0.192

Risk adjustment changes little

	Mean $\hat{\alpha}$
• Baseline	**0.192**
• Restrict to DRG 470	**0.216**
– Joint replacement without major complications, 86% of cases	

Risk adjustment changes little

	Mean $\hat{\alpha}$
• Baseline	**0.192**
• Restrict to DRG 470	**0.216**
– Joint replacement without major complications, 86% of cases	
• Use risk-adjusted surgeon quality	**0.192**
– Observed minus expected failures, with expected failures from age, sex, race, DRG, and the Charlson index	
– Correlation with raw quality is 0.84	

Risk adjustment changes little

	Mean $\hat{\alpha}$
• Baseline	**0.192**
• Restrict to DRG 470	**0.216**
– Joint replacement without major complications, 86% of cases	
• Use risk-adjusted surgeon quality	**0.192**
– Observed minus expected failures, with expected failures from age, sex, race, DRG, and the Charlson index	
– Correlation with raw quality is 0.84	

Takeaway

The estimated weight on quality is stable.

Full information: distributions across markets (myopic, $\eta = 1$)

